

[This question paper contains 4 printed pages.]

Your Roll No.....

Sr. No. of Question Paper : 2427

I

Unique Paper Code : 2343010010

Name of the Paper : Data Privacy

Name of the Course : **B.Sc. Hons Computer
Science**

Semester : V – DSE NEP-UGCF

Duration : 3 Hours

Maximum Marks : 90

Instructions for Candidates

1. Write your Roll No. on the top immediately on receipt of this question paper.
2. The Question paper consists of **two** sections.
3. **Section A** is compulsory.
4. Attempt any **Four** questions from **Section B**.
5. All parts of a question must be attempted together.

P.T.O.

Section – A

1. (a) Differentiate between Privacy and Anonymity. (3)
- (b) is the importance of public key certificates and certificate authorities? (3)
- (c) Explain the concept of the “Right to Privacy” and its relevance in the digital age. (3)
- (d) Differentiate between Insecure and Imprecise Use of Data. (3)
- (e) What are the challenges associated with privacy preservation of Time series Data? (3)
- (f) Define a replay attack. Is it an active or a passive attack? Justify your answer. (3)
- (g) What are the duties of Data Principal with respect to DPDPA. (3)
- (h) Compare and contrast Symmetric and Asymmetric encryption. (3)
- (i) How can we differentiate between Law and ethics in cybersecurity? (3)
- (j) What is the goal of privacy by design in PII? (3)

Section – B

2. (a) Explain the problems encountered by the user with the use of their personal data with the help of an example. (7)
- (b) Describe the active and passive attacks and their types. (8)
3. (a) Explain the CIA triad with respect to Information Security. Additionally, discuss how the concepts of authenticity and accountability complement these objectives. (7)
- (b) Explain the role of the following stakeholders in data privacy organization: (8)
 - (i) Customer
 - (ii) Data Analyst
 - (iii) Business operation Employee
 - (iv) Data Anonymizer
4. (a) What is personally identifiable information (PII), and why is it important to protect it in the context of data privacy and security? Which of the following examples would be considered personally identifiable information (PII)? Explain why or why not :

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- Name and address
- IP address
- Social media account information
- Anonymous survey data (7)

(b) List and briefly describe the main objectives of the GDPR. Also, explain the key responsibilities of a Data Protection Officer (DPO) in an organization. (8)

5. (a) Define Digital signatures. How are digital signatures created and verified? (7)

(b) What is the difference between Security and Privacy. What are the objectives of cybersecurity with the help of a diagram? (8)

6. (a) Briefly explain the five steps of the Model Surveillance System. (7)

(b) What are the different methods of protecting data? Explain each with the help of an example. (8)